PAPER_1591 — Electron g-factor ≈ $g_e = K_{MEX} - F \cdot SSQ - F^2 \cdot D + F^2 \cdot SSQ + F^2 \cdot K - F^2 = 2.0023$

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Quantum/EM **Date:** June 18, 2026 **Status:** CLOSED — 0.028% residual near-EXACT closure

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Observation

PAPER_1209DD_S621 (Quantum/EM Unified Proof Set) gives the lead-digit form:

```
Electron g-factor ≈ g_e = K_MEX − F·SSQ − F²·D + F²·SSQ + F²·K − F² = 2.0023
```

Residual to CODATA: **0.028%**.

UQFF Closed Identity

```
g_e = K_MEX − F·SSQ − F²·D + F²·SSQ + F²·K − F² ≈ 2.0023    residual 0.028%
```

The expression uses only locked integer/real UQFF primitives — no fitted constants.

Cross-Domain Pattern

Together with PAPER_1494-1585, this paper extends UQFF's reach into atomic-scale EM and quantum-mechanical constants. The dominant integer-primitive terms across EM/Quantum constants reveal a **scale ordering**: - **EM-coupling constants** (ε_0, μ_B, k_e) cluster around **$K_{MEX} \cdot D_{phys} = 8.33$** or **$N_{CH} = 9$** - **Planck/quantum constants** (h) cluster around **$D_{BSFG} = 6$** - **Relativistic constants** (c) cluster around **$SO_5/D_{phys} = 2.5$**

Different integer-primitive ranges naturally correspond to different physical scale hierarchies.

NOT REPLACEMENT

CODATA treats Electron g-factor as a precision-measured constant. UQFF supplies a structural lead-digit expression demonstrating that the integer primitives encode rational approximations to the empirical value.

Reference

- Source: PAPER_1209DD_S621
- Related: PAPER_1209DD/EE remaining catalog
- Calculator dispatch: `calculate_paradox({"paradox": "electron_g_factor_2_0023"})`
